



Comparison of the variability and diagnostic efficacy of respiratory-gated PET/CT based radiomics features with ungated PET/CT in lung lesions

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ABSTRACT

Objectives: To investigate the variability and diagnostic efficacy of respiratory-gated (RG) PET/CT based radiomics features compared to ungated (UG) PET/CT in the differentiation of non-small cell lung cancer (NSCLC) and benign lesions.

Methods: 117 patients with suspected lung lesions from March 2020 to May 2021 and consent to undergo UG PET/CT and chest RG PET/CT (including phase-based quiescent period gating, pQPG and phase-matched 4D PET/CT, 4DRG) were prospectively included. 377 radiomics features were extracted from PET images of each scan. Paired *t* test was used to compare UG and RG features for inter-scan variability analysis. We developed three radiomics models with UG and RG features (i.e. UGModel, pQPGModel and 4DRGModel). ROC curves were used to compare diagnostic efficiencies, and the model-level comparison of diagnostic value was performed by five-fold cross-validation. A *P* value < 0.05 was considered as statistically significant.

Results: A total of 111 patients (average age ± standard deviation was 59.1 ± 11.6 y, range, 29 – 88 y, and 63 were males) with 209 lung lesions were analyzed for features variability and the subgroup of 126 non-metastasis lesions in 91 patients without treatment before PET/CT were included for diagnosis analysis. 101/377 (26.8 %) 4DRG features and 82/377 (21.8 %) pQPG features showed significant difference compared to UG features (both *P* < 0.05). 61/377 (16.2 %) and 59/377 (15.6 %) of them showed significantly better discriminant ability (ΔAUC % (i.e. $(AUC_{RG} - AUC_{UG}) / AUC_{UG} \times 100\%$) > 0 and *P* < 0.05) in malignant recognition, respectively. For the model-level comparison, 4DRGModel achieved the highest diagnostic efficacy (sen 73.2 %, spe 87.3 %) compared with UGModel (sen 57.7 %, spe 76.4 %) and pQPGModel (sen 63.4 %, spe 81.8 %).

Conclusion: RG PET/CT performs better in the quantitative assessment of metabolic heterogeneity for lung lesions and the subsequent diagnosis in patients with NSCLC compared with UG PET/CT.

1. Introduction

Lung cancer is still a major global burden of cancer, with an estimated incidence of 2.2 million new cases and 1.8 million cancer deaths worldwide in 2020 [1,2]. It ranks first in the cancer-related deaths and contributes to 30 % of all cancer deaths in China with increasing

mortality rate and incidence [3]. ¹⁸F-FDG PET/CT plays a crucial role in the clinical assessment and management of lung cancer, offering the benefit of superior sensitivity in characterizing and quantifying the biological landscape of lesions compared to other tomographic imaging modalities. Therefore, it is strongly recommended for further evaluation of patients with a malignant risk of > 10 % (Brock model) when

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available [4]. Yet, considering that PET images acquisitions last several minutes, the influence of respiratory motion in the thorax may lead to the image blurring of PET and misalignments between CT and PET images. These issues contribute to significant image distortion and quantitative errors when assessing tumor morphology and uptake activity in PET/CT images, potentially resulting in clinical misinterpretation. Therefore, efficient methods for correcting respiratory motion are in need to tackle these challenges.

Over the past years, different respiratory gating techniques have been developed for respiratory motion correction, such as quiescent period gating (QPG) techniques (amplitude- and phase-based gating) and respiratory-gated 4D PET/CT. Focusing on correcting respiratory motion artifacts in PET images, QPG techniques mitigate the impact of respiratory motion by extracting data during the relative quiescent state of the patient's respiration for reconstruction. Compared to amplitude-based QPG (aQPG), phase-based QPG (pQPG) could compensate for patient respiratory curve baseline drift which happens frequently in PET imaging. However, these techniques didn't account for the CT component [5] and come at the expense of an extended scan time [6,7]. In contrast, novel 4D PET/CT aims to merge different phases of PET series with modern data-driven methods (e.g. elastic transformation based on the approach of optical flow), while simultaneously attenuation-correcting the different phases of PET series with the phase-matched low-dose CT. In this way, 4D PET/CT delivers high-quality gated images corrected for respiratory motion without a substantial increase in acquisition time [8]. Although established respiratory-gated PET/CT has a role in lesion detection and characterization, it is underutilized due to practical limitations and an unclear patient management pathway.

However, the diagnostic value of PET/CT was still restricted by the intrinsic characteristic of standard uptake value (SUV) as a semi-quantitative metabolic parameter, which is influenced by a variety of factors, such as scan time, plasma glucose, reconstruction methods, region of interest (ROI), etc. With the development of medical image analysis techniques, PET/CT radiomics analysis, which could quantify the intrinsic spatial and temporal heterogeneity, has shown promising results for reducing false positive rate [9] and improving precision [10]. Hence, radiomics features have been regarded as important indices for the accurate diagnosis of lung lesions. However, the impact of respiratory motion and the incremental value of RG PET/CT in radiomics features have not been well studied, which limited the comprehensive evaluation of the diagnostic value of RG PET/CT.

Therefore, the purpose of this study was to assess the incremental value of RG PET/CT in quantitative characterization and diagnosis of lung lesions by investigating the variability as well as diagnostic efficacy of RG PET/CT based radiomics features compared to ungated PET/CT.

2. Material and methods

2.1. Dataset

This study was approved by the ethics committee of Cancer Hospital & Shenzhen Hospital, Chinese Academy of Medical Sciences (2020–36). A total of 117 patients with suspected lung lesions from March 2020 to May 2021 were prospectively included in this study. All patients underwent ungated (UG) PET/CT and chest respiratory-gated (RG) PET/CT. Exclusion criteria were as following: 1) Failed RG PET/CT (usually due to irregular respiratory curve, patient nervousness etc.); 2) Lesions without avid FDG uptake; 3) Lesions that were diffuse distributed (i.e. > 10) and difficult for precise segmentation; 4) Lesions detected only by UG or RG PET/CT. 6 patients failed RG PET/CT. 209 lung lesions in 111 cases were analyzed for variability of features (not including 5 lesions detected only by RG PET/CT). Exclusions for diagnosis analysis were as following: 1) Treatment before PET/CT (e.g. Chemoradiotherapy, immunotherapy, targeted therapy for malignancies, and anti-inflammatory therapy for benign lesions); 2) Pulmonary metastasis; 3) Other pulmonary malignancy except NSCLC confirmed either before or

after PET/CT scan; 4) Lost to follow-up. Finally, the subgroup of 126 lesions in 91 patients were included in diagnosis analysis, including 55 benign lesions of 27 cases and 71 malignant lesions of 64 cases. The flow chart of enrollment was shown in Fig. 1.

2.2. PET/CT protocol

All patients fasted for at least 6 h before the PET/CT examination (Discovery MI PET/CT, GE Healthcare, Milwaukee, USA), and their blood glucose levels were ≤ 8.0 mmol/L before ^{18}F -FDG injection (4.44–5.55 MBq/kg). The patients first underwent a whole-body CT scan (120 kV, automatic tube current modulation range from 10 to 220 mA, pitch of 1.375:1 and noise index of 20) from the top of skull to the upper thighs. Whole-body PET scanning was then performed immediately at six to eight bed positions (2 min/bed). The patients were scanned with the arms raised during the whole examination.

For the chest RG PET/CT (5 min/bed), respiratory gating was performed using real-time position management system (RPM; Varian Medical Systems, California, USA). The system uses an infrared-reflective marker block that is placed on each patient's chest and an infrared camera fixed on the foot-side edge of the PET/CT table to continuously track the respiratory motion and activated gated data acquisition. The phase-based quiescent period gating (pQPG; Q.Static, GE Healthcare, USA) is an established phase-based gating reconstruction (only extract data during the relative quiescent phase, usually refer to the end expiration breath) integrated into the clinical whole-body PET protocol, which is available simultaneously with the conventional static PET images without the need for additional CT radiation dose. Patients then underwent 4D respired-gated PET/CT (4DRG; Q.Freeze, GE Healthcare, USA) using 100 % PET counts of total respiratory cycle in combination with a phase-matched low dose cine CT (100 kV, 100–180 mA) to provide high-quality spatial alignment and attenuation correction.

2.3. Radiomics analysis

The original PET images were initially resampled to standardize the pixel spacing. All lung lesions were annotated manually using ITK-SNAP by two experienced nuclear medicine physicians in UG, pQPG, and 4DRG PET images, carefully segmenting the borders between lung lesions and hilum lymph nodes, pulmonary atelectasis, and chest wall. Final consensus was reached after comprehensive image analysis without knowing the pathological results. The lesions were randomly distributed in training and validation sets with the proportion of 8:2 approximately. For each lesion, 377 radiomics features were extracted from PET images of each scan using the PyRadiomics library, including 71 histogram-related features, 89 shape-related features, 62 textural features and 155 wavelet features. The analysis was performed in accordance with the most current baseline values of the Image Biomarker Standardization Initiative (IBSI). Considering the large number of features extracted, all features of different scanning methods (UG, pQPG and 4DRG) were screened by high variance, *t*-test, minimum redundancy maximum correlation method and least absolute shrinkage and selection operator (LASSO) in turn. For a certain feature, the greater the AUC value and internal variance of all samples, the more significant the difference between benign and malignant samples. Besides, maximizing the correlation between features and categorical variables, while minimizing the correlation between features can help to reduce unnecessary features. Finally, multiple logistic regression analysis was applied to analyze the samples under the three scanning modes, and the generalized linear model (GLM) was established with the selected features of UG (i.e. UGModel) and RG (i.e. pQPGModel and 4DRGModel).

2.4. Statistical analysis

The data was presented as mean $\pm$ standard deviation for

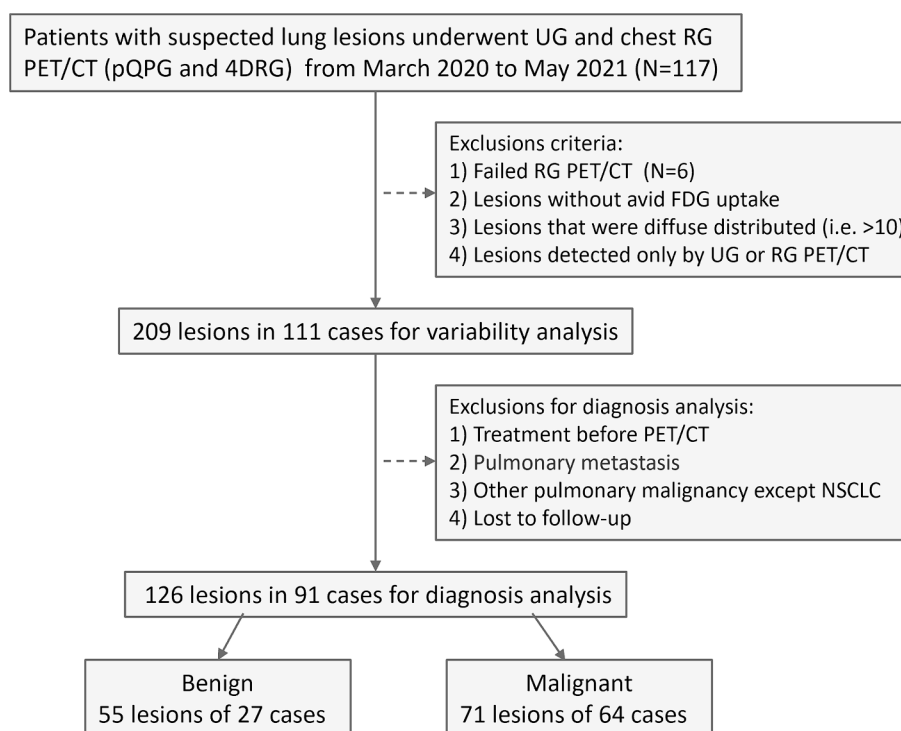


Fig. 1. The enrollment flow chart of patients with suspected lung lesions.

metabolism-related parameters, such as maximum standardized uptake value (SUVmax), mean standardized uptake value (SUVmean), metabolic tumor volume (MTV), and total lesion glycolysis (TLG), and were compared with a paired *t* test. While the independent *t*-test and analysis of variance (ANOVA) were applied to the lesion stratification analysis of metabolism-related parameters variation between RG and UG. The feature variability and diagnostic value of RG and UG scans were compared via paired *t*-test and receiver operating characteristic (ROC) curves respectively. The model-level comparison was performed using five-fold cross-validation. The validation sets from each of the five cross-validations were consolidated into a new dataset, and individual performance metrics were calculated for this new dataset. The diagnostic efficacies of different models, including sensitivity, specificity, accuracy, positive predictive value (PPV), and negative predictive value (NPV), were compared using the McNemar's chi-squared test (DTCComPair package) and AUC of ROC curves (pROC package) on Rstudio platform (v4.1.3). Matlab (R2020b) and Graphpad Prism (v9.1.0) were used for *t*-test and graphic analysis. A *P* value < 0.05 was considered as statistically significant.

3. Results

3.1. General characteristics

The general characteristics of all 214 lung lesions in 111 cases were presented in Table 1 (detailed in Supplementary Table 1). The average age $\pm$ standard deviation of all patients was 59.1 ± 11.6 y (range, 29 – 88 y), and 56.8 % of which (63/111) were males. Follow-up was carried out from March 2020 to December 2022. The reference standard for diagnosis was based on histology (either biopsy or surgery) and/or long-term clinical follow-up (>6 months). 126 lung lesions in 91 patients were included in diagnosis analysis, which were confirmed by histology (either biopsy or operation) for 89 (70.6 %) lesions and follow-up for 37 (29.4 %) lesions. For the lesions-based diagnosis analysis, 71 malignant lesions of 64 cases consisted of 54 adenocarcinoma (76.1 %), 16 squamous cell carcinoma (22.5 %) and 1 lung cancer (unspecified), while 55

Table 1
General characteristics of dataset.

	Case-based (%)	Lesion-based (%)
Total	111	214
Age	59.1 ± 11.6 y (29 – 88 y)	–
Male	63 (56.8)	–
Location of lesions (lobe)		
right upper	–	60 (28.0)
right middle	–	14 (6.5)
right lower	–	39 (18.2)
left upper	–	67 (32.2)
left lower	–	32 (15.9)
Involved both upper and lower lobes*	–	2
Size (cm)		
≤ 1	–	64 (29.9)
$1 < D \leq 3$	–	96 (44.9)
$3 < D \leq 5$	–	40 (18.7)
> 5	–	14 (6.5)
Diagnosis analysis	91	126
histology	–	89 (70.6)
follow-up	–	37 (29.4)
Malignant	64 (70.3)	71 (56.3)
adenocarcinoma	–	54 (76.1)
squamous cell carcinoma	–	16 (22.5)
lung cancer (unspecified)	–	1 (1.4)
Benign	27 (29.7)	55 (43.7)
inflammation	–	38 (69.1)
tuberculosis	–	4 (7.3)
cryptococcal pneumonia	–	3 (5.5)
inflammatory myofibroblastic tumor	–	1 (1.8)
sclerosing alveolar cell tumor	–	1 (1.8)
neurilemmoma	–	1 (1.8)
other benign lesions	–	7 (12.7)

* Two lesions grew across left upper and lower lobes.

benign lesions of 27 cases included 38 inflammation (69.1 %), 4 tuberculosis (7.3 %), 3 cryptococcal pneumonia (5.5 %), 1 inflammatory myofibroblastic tumor, 1 sclerosing alveolar cell tumor, 1 neurilemmoma (1.8 %), and 7 other benign lesions (12.7 %).

3.2. Comparisons of metabolism-related parameters

For 209 lesions in 111 cases detected by both UG and RG PET/CT, metabolism-related parameters were measured in the picture archiving and communication system (PACS), including maximum and mean standardized uptake value (SUVmax and SUVmean), metabolic tumor volume (MTV), and total lesion glycolysis (TLG). MTV was measured as the metabolic volume of SUV over 40 % SUVmax of the lesion. And $TLG = SUV_{mean} \times MTV$. The SUVmax of UG, pQPG and 4DRG PET/CT were 7.62 ± 6.19 , 7.56 ± 6.07 and 8.57 ± 6.75 ; SUVmean were 4.60 ± 3.73 , 4.63 ± 3.70 and 5.35 ± 4.17 ; and TLG were 58.02 ± 173.78 , 58.14 ± 172.76 and 64.67 ± 191.27 , respectively (all P values of 4DRG vs. UG and pQPG < 0.05; P values of pQPG vs. UG > 0.05). The MTV of UG, pQPG and 4DRG were similar (6.90 ± 14.38 , 7.00 ± 14.57 and 6.85 ± 14.08 , all P > 0.05) (Fig. 2).

We further stratified the 209 lesions with size (longest diameter measured in CT), location (upper/middle and lower lobe) and metabolic activity (SUVmax measured in UG PET), and calculated the variation of metabolism-related parameters between RG and UG (i.e. $(RG - UG) / UG \times 100\%$). Overall, SUVmax and SUVmean of 4DRG increased by 17.74 % and 21.98 % compared to UG, with the higher $\Delta SUV\%$ in the groups of lesions ≤ 1 cm (ΔSUV_{max} 26.37 %, $P=0.070$; ΔSUV_{mean} 33.70 %, $P=0.048$), located in lower lobe (ΔSUV_{max} 30.76 % vs. 11.35 %, $P<0.001$; ΔSUV_{mean} 35.27 % vs. 15.50 %, $P<0.001$), and $UG-SUV_{max} \leq 2.5$ (ΔSUV_{max} 31.07 % vs. 13.65 %, $P=0.002$; ΔSUV_{mean} 34.25 % vs. 18.22 %, $P=0.006$). Moreover, lesions ≤ 1 cm have significantly higher ΔMTV (38.19 %, $P=0.009$) and ΔTLG (47.43 %, $P=0.005$). As for pQPG vs. UG, ΔMTV and ΔTLG showed similar tendencies to that of 4DRG, but no significant differences were observed in all subgroups (Table 2, Fig. 3). Representative images of UG and RG PET/CT were shown in Fig. 4a,b. Another notable point is that there were 5 lesions (all of them with diameter < 1 cm and $UG-SUV_{max} \leq 2.5$) detected only by 4DRG PET as FDG-avid, also suggesting that respiratory gating may partially compensate the influence of partial volume effect and improve the detection of small and low metabolic lesions (Fig. 4c).

3.3. Feature-level comparison of variability and discriminant ability

The 377 radiomics features were extracted from PET images of each scan were detailed in Supplementary Table 2. For the feature-level comparison, 101/377 (26.8 %) 4DRG and 82/377 (21.8 %) pQPG features showed significant difference compared to UG features (both $P<0.05$, area I and II in Fig. 5a,b). Compared to UG PET, 4DRG caused significant differences in all four kinds of image quantitative features, specifically 53.2 % (33/62) textural features and 42.3 % (30/71) histogram-related features. While pQPG and UG PET were mainly different in shape-related features (44.9 %, 40/89) and histogram-related features (28.2 %, 20/71) (Fig. 5c).

As for the 126 lesions in 91 patients included in diagnosis analysis, the AUCs of 229/377 (60.7 %) 4DRG and 133/377 (35.3 %) pQPG features were preferable to that of UG features in malignant recognition

($\Delta AUC\% > 0$, area II and III in Fig. 5a,b). Among of which, 61/377 (16.2 %) 4DRG and 59/377 (15.6 %) pQPG features had significantly incremental diagnostic value ($\Delta AUC\% > 0$ and $P < 0.05$, area II).

3.4. Diagnostic value of different models

We selected and developed three radiomics models for the discrimination of NSCLC and benign lung lesions, including 19 UG, 20 pQPG and 14 4DRG features respectively (Supplementary Table 2). Among of which, 30 % (6/20) and 64.3 % (9/14) features in pQPGModel and 4DRGModel had significantly incremental diagnostic value than that of UGModel respectively ($\Delta AUC\% > 0$ and $P < 0.05$).

For the model-level comparison of diagnostic efficiencies, the AUC, accuracy, sensitivity, specificity, PPV and NPV of the three models were 0.75, 65.9 %, 57.7 %, 76.4 %, 75.9 %, 58.3 % for UGModel, and 0.79, 71.4 %, 63.4 %, 81.8 %, 81.8 %, 63.4 % for pQPGModel, and 0.80, 79.4 %, 73.2 %, 87.3 %, 88.1 %, 71.6 % for 4DRGModel, respectively (Table 3). pQPGModel corrected 14.3 % (18/126) UGModel diagnoses, identifying 8 malignant and 10 benign lesions with the opposite diagnosis on UGModel. While 4DRGModel changed 19.8 % (25/126, 14 malignant and 11 benign lesions) diagnoses of UGModel (Table 4), and achieved the highest AUC of 0.80 (0.75 for UGModel and 0.79 for pQPGModel) (Fig. 6).

4. Discussion

With the known adverse effects of respiratory motion artifacts on image quality and quantitative analysis, an important question remains whether the use of respiratory gating results in improved heterogeneity characterization and diagnosis of lung lesions. To assess the impact of respiratory motion artifact and the incremental value of RG PET/CT in radiomics features is important in evaluating the diagnostic benefit of RG PET/CT for lung lesions. In this prospective study, we provided a comprehensive lesion-based analysis comparing the variability and diagnostic efficacy of 377 IBSI radiomics features of different respiratory-gated techniques (pQPG and 4DRG) with ungated PET/CT. Our results indicate that RG PET/CT, especially 4D PET/CT, improves not only image quality but also quantitative assessment of metabolic heterogeneity for lung lesions, and subsequently help improving the diagnosis of NSCLC.

Respiratory gating PET/CT technique allows a more accurate quantitation of metabolic activity and definition of the lesion volume which are distorted because of the breathing motion. Early studies on respiratory gating PET/CT mainly focused on the enhancement of semi-quantitative metabolic parameters. The SUVmax of RG increased by 2.2 % to 89.9 % as reported in previous studies [11–14], and the MTV decreased by 1.3 % to 34 % [14–17] compared to UG PET/CT. The large range of variation may be related to the gating methods (amplitude or phase-based gating, 4D PET/CT), image reconstruction methods, measurement methods (automatic and manual segmentation, MTV threshold), and patient selection etc. According to our results, although

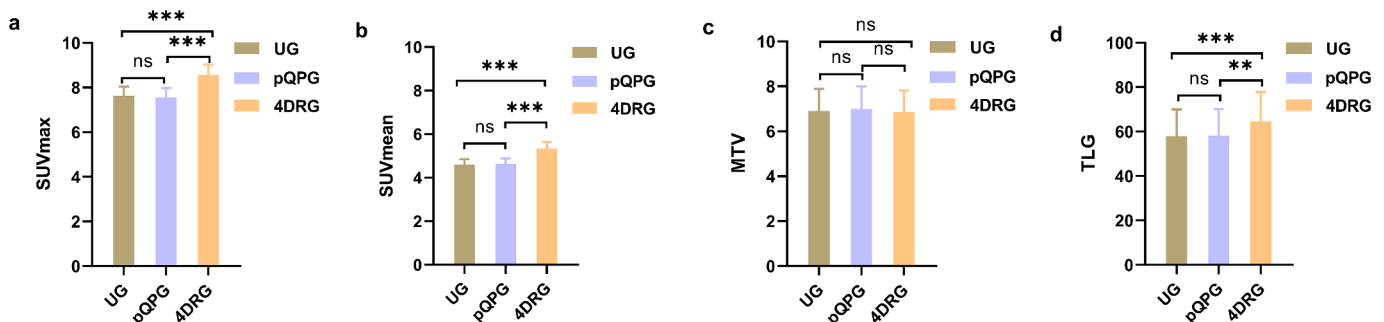
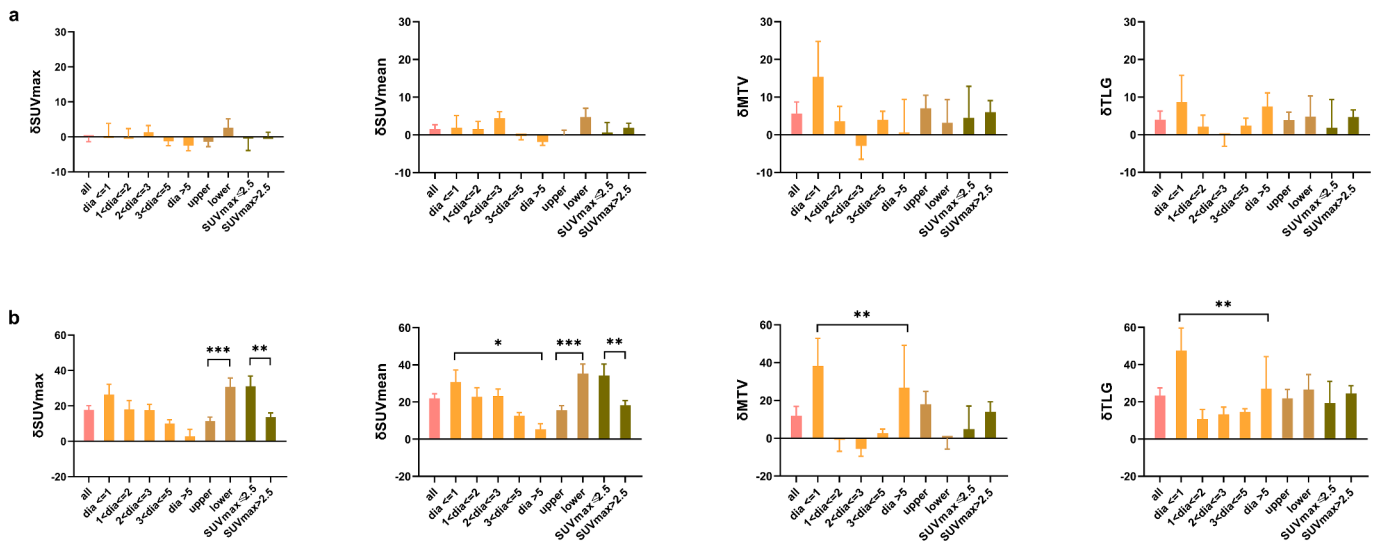


Fig. 2. Comparisons of metabolism-related parameters of UG and RG.

Table 2

Stratified comparisons of metabolism-related parameters variation of RG and UG.

	δ SUVmax (%)	P	δ SUVmean (%)	P	δ MTV (%)	P	δ TLG (%)	P
pQPG	-0.09 ± 18.23		1.56 ± 16.58		5.66 ± 44.16		4.00 ± 33.04	
Diameter (cm)								
D ≤ 1	0.29 ± 27.24	0.957	1.87 ± 25.16	0.710	15.34 ± 72.36	0.336	8.67 ± 54.56	0.705
1 < D ≤ 2	0.07 ± 17.63		1.60 ± 15.48		3.61 ± 30.52		2.13 ± 23.70	
2 < D ≤ 3	1.31 ± 11.62		4.38 ± 10.86		-2.96 ± 21.41		-0.18 ± 17.51	
3 < D ≤ 5	-1.29 ± 7.85		-0.28 ± 6.14		3.98 ± 14.43		2.46 ± 12.25	
5 < D	-2.49 ± 5.45		-1.89 ± 3.22		0.65 ± 32.79		7.50 ± 13.52	
Location (lobe)								
Upper/ middle	-1.38 ± 16.53	0.141	0.02 ± 14.73	0.055	7.01 ± 40.71	0.562	3.88 ± 24.76	0.847
Lower	2.60 ± 21.30		4.72 ± 19.71		3.20 ± 51.13		4.83 ± 45.60	
UG-SUVmax								
≤ 2.5	-0.48 ± 23.87	0.863	0.65 ± 18.48	0.661	4.55 ± 58.39	0.841	1.78 ± 53.06	0.591
> 2.5	0.04 ± 16.20		1.84 ± 16.01		6.00 ± 39.01		4.68 ± 24.01	
4DRG	17.74 ± 33.90		21.98 ± 35.64		11.89 ± 71.33		23.37 ± 59.27	
Diameter (cm)								
D ≤ 1	26.37 ± 44.80	0.070	30.77 ± 49.42	0.048*	38.19 ± 113.24	0.009**	47.43 ± 93.63	0.005**
1 < D ≤ 2	17.96 ± 38.77		22.74 ± 38.31		-0.84 ± 46.65		10.76 ± 40.02	
2 < D ≤ 3	17.56 ± 20.15		23.22 ± 22.37		-5.56 ± 23.69		13.27 ± 23.20	
3 < D ≤ 5	10.03 ± 13.43		12.59 ± 11.09		2.69 ± 13.84		14.59 ± 10.88	
5 < D	2.85 ± 14.69		5.30 ± 11.14		26.75 ± 83.78		27.11 ± 64.21	
Location (lobe)								
Upper/ middle	11.35 ± 27.58	<0.001***	15.50 ± 30.13	<0.001***	18.00 ± 80.67	0.089	21.92 ± 56.03	0.593
Lower	30.76 ± 41.55		35.27 ± 42.29		-0.01 ± 47.23		26.63 ± 66.33	
UG-SUVmax								
≤ 2.5	31.07 ± 40.73	0.002**	34.25 ± 43.36	0.006**	4.81 ± 85.88	0.429	19.37 ± 81.24	0.590
> 2.5	13.65 ± 30.51		18.22 ± 32.15		14.05 ± 66.41		24.60 ± 50.96	

Statistical significance: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$, the same below.**Fig. 3.** Stratified comparisons of metabolism-related parameters variation of (a) pQPG vs. UG and (b) 4DRG vs. UG.

pQPG PET/CT could help improving the image quality, no significant differences were observed in metabolism-related parameters between UG and pQPG PET/CT, which may be the result of a partial count loss in respiratory cycle. Some researchers suggested that the amplitude-based QPG PET/CT was superior to phase-based QPG in patients with irregular breathing patterns [18] and shows consistency in improving accuracy in quantitative analysis similar to 4D PET/CT [19]. As for 4D PET/CT, SUVs were significantly higher than that of UG PET/CT with an average increase of SUVmax by 17.74 % and SUVmean by 21.98 %. And the extent of increase in SUVs was observed to a greater extent in lesions of lower lobes, FDG hypometabolism (UG-SUVmax ≤ 2.5) and inversely proportional to the lesion size, because the error associated with activity estimation is related to degree of lesion motion [12,15] and partial volume effect [20]. Despite the disputed results, the MTV of UG, pQPG

and 4DRG has no statistical difference similar to some studies [19,21], indicating that the measurement of MTV with a relative threshold of 40 % SUVmax might be less affected by respiration artifact. However, Δ MTV% of 4DRG tend to increase in sub-centimeter (≤ 1 cm) or large lesions (> 5 cm), probably due to the significant increase of SUVs for correction of partial volume effect in small lesions and increased inhomogeneity uptake within the large lesions. Similar trend was observed in Δ TLG%. Since MTV and TLG were reported to play an important role in patient risk stratification and prognosis prediction [22], the measuring method and errors caused by respiratory motion should be noted in related studies.

Apart from the improved image quality and quantitative accuracy, reduced motion blurring by RG PET/CT also has an important impact in delineating tumoral heterogeneity (intrinsic FDG accumulation).

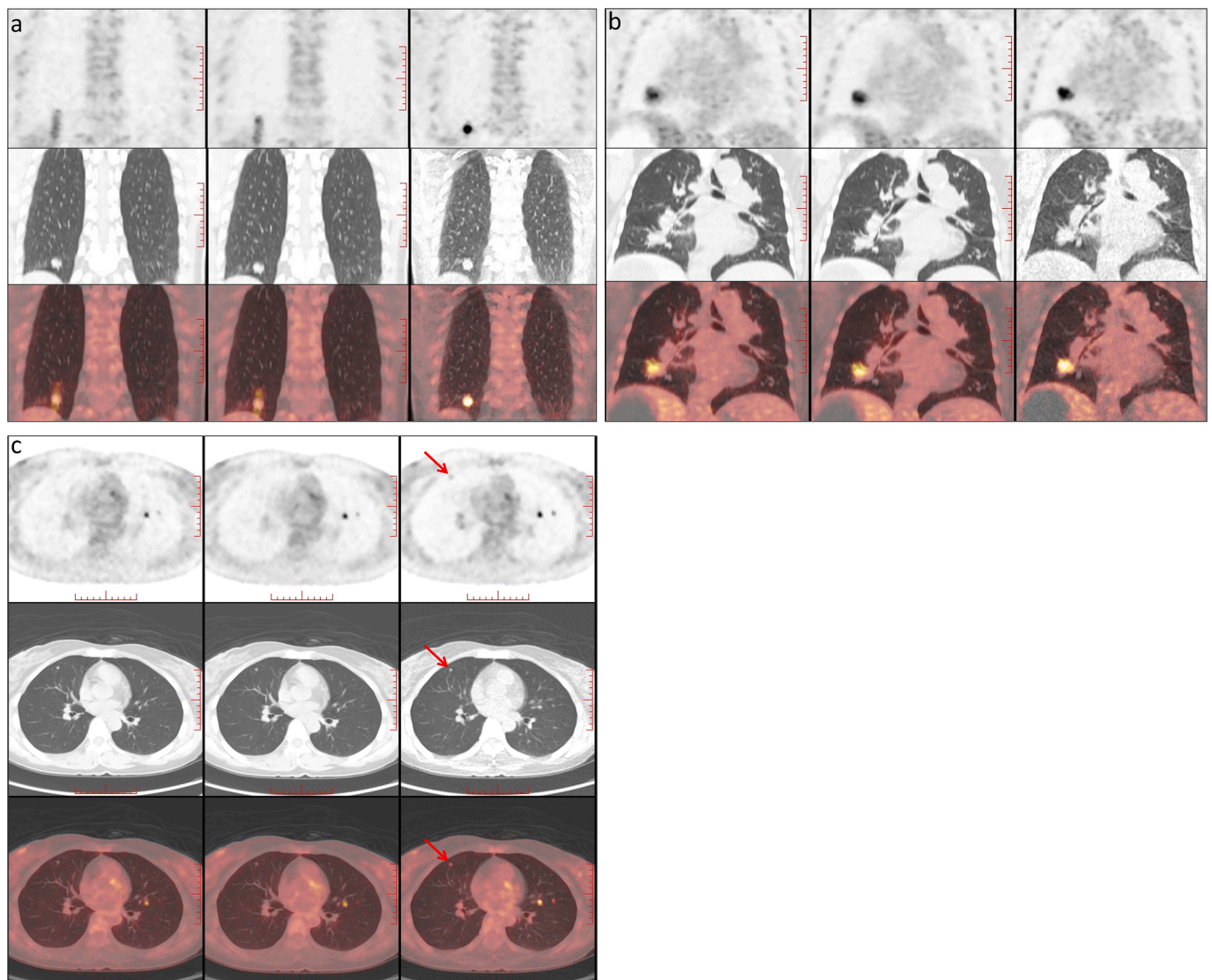


Fig. 4. Representative images of UG, pQPG and 4DRG PET/CT: (a) A 48-year-old male with low differentiated adenocarcinoma of lower lobe of right lung. RG images especially the 4DRG significantly improved not only the image quality and matching, but also the metabolic distribution and quantification compared to UG (SUVmax of UG, pQPG and 4DRG were 4.6, 4.8 and 12.6, respectively; MTV were 3.32, 1.23 and 1.15, respectively). (b) UGModel false negative (FN), pQPGModel and 4DRGModel true positive (TP): A 64-year-old female with adenocarcinoma of lower lobe of right lung. The SUVmax and MTV were similar between pQPG and UG PET/CT. For 4DRG, this lesion showed a 45.8% increase in the SUVmax, and a 14.3% reduction in MTV compared to UG (SUVmax of UG, pQPG and 4DRG were 7.2, 7.6 and 10.5, respectively; MTV were 6.99, 7.01 and 5.99, respectively). And the diagnosis of pQPGModel and 4DRGModel were both correct based on their improved quantitative features. (c) Lesion detected only by 4DRG PET as FDG-avid: A 42-year-old female diagnosed as adenocarcinoma of upper lobe of right lung. And a small subpleural metastatic lesion that were detected only by 4DRG PET as FDG-avid (SUVmax 1.2).

Previous studies with relatively small numbers of patients investigated a significant change in a number of radiomic features of RG PET/CT compared to UG PET/CT. Oliver et al. [23] assessed 56 features from 23 lung cancer patients consisting of shape descriptors, first- and second-order texture features, and found only 26.2 % of total features demonstrated less than 5 % difference between 3D and RG protocols for PET. We investigated the variability of 377 IBSI radiomics features, which to our best known was the largest number of radiomics features of respiratory-gated PET/CT compared to previous studies (4–56 features) [10,23–26]. According to our results of variability analysis of 377 features, RG PET/CT had more than 20 % variable features compared to UG PET/CT (26.8 % of 4DRG, and 21.8 % of pQPG). Both pQPG and 4DRG caused significant differences in histogram-related features which mainly measure the intensity of the image gray value. Besides, pQPG technique only extract data during the relative quiescent phase (usually refer to the end expiration breath) to mitigate the blurring and distortion

caused by respiratory motion, therefore can also help improve quantification of shape-related features. With the advantage of not adding extra scanning time and radiation dose, pQPG might be more favorable to most patient with lung lesions and clinical practice. Moreover, 4DRG with phase-matched CT caused significant differences especially in textural features, which may have higher value in improving quantitative analysis and heterogeneity characterization. And the 4DRG PET/CT based features might be more reliable for radiomic analysis.

Preliminary studies suggested improved lesion detection and diagnosis with respiratory gated PET/CT. Vicente et al. [27] reported that 4D PET/CT change 40 % classification of cases from benign to malignant in 42 hypometabolic pulmonary lesions (UG-SUVmax < 2.5) of 28 patients. Guerra et al. [16] retrospectively analyzed 206 lung lesions in 155 patients from 5 European centers and found that 4D PET/CT improved quantification and confidence in reporting, reducing undetermined findings in UG PET/CT and increasing the overall accuracy

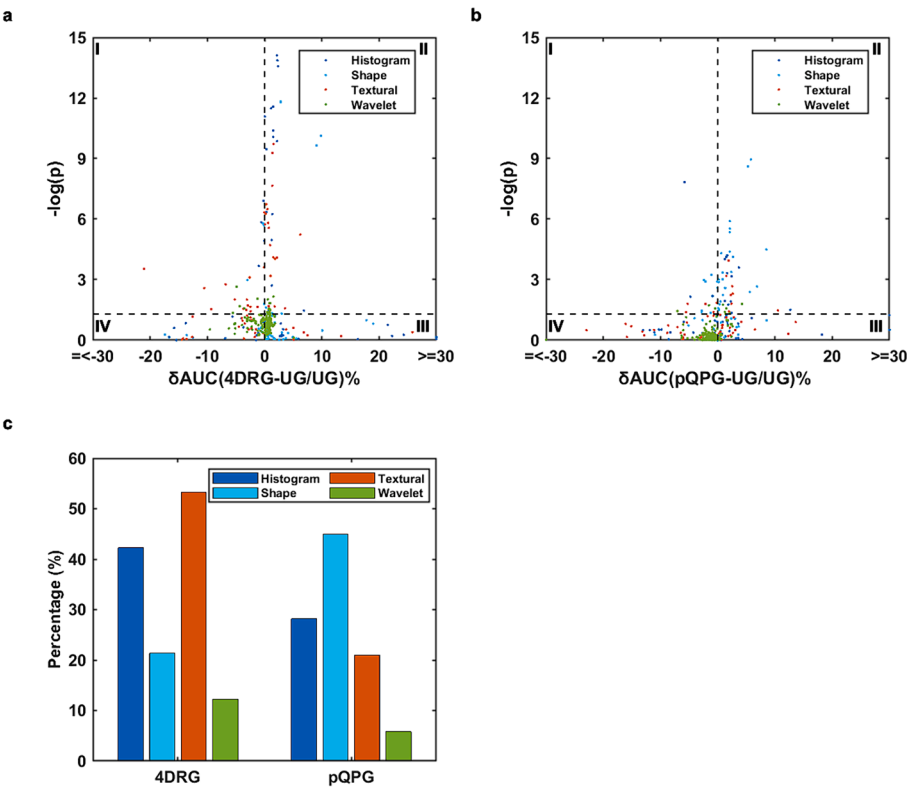


Fig. 5. Feature-level comparison of variability and discriminant ability of 377 features: (a) 4DRG compared to UG PET/CT (I: 40, II: 61, III: 168, IV: 89, $\delta\text{AUC}=0$: 19) and (b) pQPG compared to UG PET/CT (I: 23, II: 59, III: 74, IV: 193, $\delta\text{AUC}=0$: 28). (c) Type distribution of variable features of RG compared to UG.

Table 3
Diagnostic efficiencies comparison of different models.

	AUC	Accuracy(%)	Sensitivity(%)	Specificity(%)	PPV(%)	NPV(%)
UGModel	0.75 (0.67–0.84)	65.9 (56.8–74.0)	57.7 (46.3–69.2)	76.4 (61.0–84.5)	75.9 (64.5–87.3)	58.3 (46.9–69.7)
pQPGModel	0.79 (0.71–0.87)	71.4 (62.8–80.1)	63.4 (52.2–74.6)	81.8 (71.6–92.0)	81.8 (71.6–92.0)	63.4 (52.2–74.6)
4DRGModel	0.80 (0.72–0.88)	79.4 (71.1–85.9)	73.2 (62.9–83.5)	87.3 (78.5–96.1)	88.1 (79.9–96.4)	71.6 (60.8–82.4)
P value [†]						
pQPGModel	0.415	0.145	0.248	0.317	0.196	0.134
4DRGModel	0.084	0.024*	0.005**	0.034*	0.006**	0.0005***

[†] Statistical significance compared to UGModel.

Table 4
Cross-tabulated diagnosis of UG, pQPG and 4DRG models.

	Malignant		Benign	
	UG positive	UG negative	UG positive	UG negative
pQPG positive	39	8	5	2
pQPG negative	4	20	10	38
4DRG positive	41	14	4	6
4DRG negative	2	14	11	34

(94.2 % vs. 80.5 %). To evaluate the diagnostic value of RG PET/CT, we developed three radiomics models with UG and RG features and found that RG features-based models especially 4DRGModel showed better diagnostic value than UGModel. pQPGModel improved sensitivity from 57.7 % to 63.4 % ($P=0.248$) and specificity from 76.4 % to 81.8 %

($P=0.317$) compared with UGModel. Moreover, 4DRGModel significantly improved diagnostic efficiencies especially in terms of sensitivity (57.7 % to 73.2 %, $P=0.005$) and NPV (58.3 % to 71.6 %, $P=0.0005$), helping reduce missed diagnosed cases. RG PET/CT based models corrected 14.3 % to 19.8 % of UGModel diagnoses. Given that 30 % (6/20) and 64.3 % (9/14) features in pQPGModel and 4DRGModel respectively had significantly incremental diagnostic value than that of UGModel, we suggested that the improved quantitative features from RG PET/CT contributes to the improvement of model performance in the diagnosis of NSCLC. When measuring tumor heterogeneity characteristics, reduced motion blurring by RG PET/CT offers a better spatial resolution of textural features which help improve diagnosis and management of lung cancer patients.

A major limitation of the current study was the relatively small amount of patient collection from a single center, which may lead to selection bias. However, we provided comprehensive lesion-based analysis of variability and diagnostic value of vast radiomics features. Secondly, we mainly divided lesions manually into upper/middle and lower lobes according to their anatomical structure, which may not

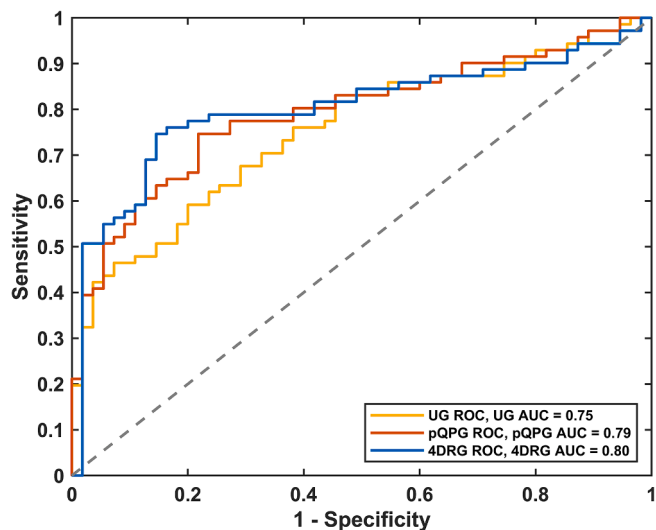


Fig. 6. The ROC curves of different radiomics features-based models.

reflect the precise location characteristics affected by respiration motion in some cases. Lastly, although 4DRG PET (sen 73.2 %, spe 87.3 %) based radiomics model showed better diagnostic efficacies than UG (sen 57.7 %, spe 76.4 %) and pQPG PET (sen 63.4 %, spe 81.1 %) according to our results, we didn't include CT features and other clinical signatures (such as serum tumor marker, tobacco exposure history and so on), which may limit the diagnostic efficacies. With the development of new respiratory gating techniques, such as data-driven gating (DDG) [28] and software-gating [29], RG PET/CT is promising to improve detection and assessment of lung lesions, providing functional information and improving the definition of the tumor used for surgical and radio-therapeutic planning. Moreover, respiratory-gated PET/CT based radiomics and deep learning features allow more accurate assessment of tumor heterogeneity and may help application of PET/CT based radiomics features for histological and molecular classification, patient stratification, therapeutic and prognostic prediction of lung cancers [30]. Further systemic advantage analysis with larger sample size is still required to help to define more clearly the clinical value of RG PET/CT in lung lesions.

5. Conclusions

Respiratory-gated PET/CT improves image quality and quantitative assessment of metabolic heterogeneity of lung lesions by reducing respiratory motion-related artifacts. Compared to ungated PET/CT, RG PET/CT especially the 4D PET/CT based radiomics features showed incremental diagnostic value in both feature-level and model-level for differentiation of NSCLC and benign lung lesions, suggesting its potential incremental value in clinical application and patient management.

CRedit authorship contribution statement

Shengyun Huang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Caifang Cao:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Linna Guo:** Visualization, Investigation, Formal analysis, Data curation. **Chengze Li:** Methodology, Formal analysis, Data curation. **Feng Zhang:** Methodology, Formal analysis, Data curation. **Yiluo Li:** Investigation, Data curation. **Ying Liang:** Writing – review & editing, Conceptualization. **Wei Mu:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lungcan.2024.107889>.

References

- [1] H. Sung, J. Ferlay, R.L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, et al., Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries, *CA Cancer J. Clin.* 71 (2021) 209–249, <https://doi.org/10.3322/caac.21660>.
- [2] J.M. Kocarnik, K. Compton, F.E. Dean, W. Fu, B.L. Gaw, J.D. Harvey, et al., Cancer incidence, mortality, years of life lost, years lived with disability, and disability-adjusted life years for 29 cancer groups from 2010 to 2019: a systematic analysis for the global burden of disease study 2019, *JAMA Oncol.* 8 (2022) 420–444, <https://doi.org/10.1001/jamaoncol.2021.6987>.
- [3] D. Yang, Y. Liu, C. Bai, X. Wang, C.A. Powell, Epidemiology of lung cancer and lung cancer screening programs in China and the United States, *Cancer Lett.* 468 (2020) 82–87, <https://doi.org/10.1016/j.canlet.2019.10.009>.
- [4] M.E. Callister, D.R. Baldwin, A.R. Akram, S. Barnard, P. Cane, J. Draffan, et al., British thoracic society guidelines for the investigation and management of pulmonary nodules. *Thorax*. 2015;70 Suppl 2:ii1-ii54. <https://doi.org/10.1136/thoraxjnl-2015-207168>.
- [5] W. Grootjans, D.D.D. Rietbergen, F.H.P. van Velden, Added value of respiratory gating in positron emission tomography for the clinical management of lung cancer patients, *Semin. Nucl. Med.* 52 (2022) 745–758, <https://doi.org/10.1053/j.semnuclmed.2022.04.006>.
- [6] C. Liu, A. Alessio, L. Pierce, K. Thielemans, S. Wollenweber, A. Ganin, et al., Quiescent period respiratory gating for PET/CT, *Med. Phys.* 37 (2010) 5037–5043, <https://doi.org/10.1118/1.3480508>.
- [7] A. Pépin, J. Daouk, P. Bailly, S. Hapdey, M.E. Meyer, Management of respiratory motion in PET/computed tomography: the state of the art, *Nucl. Med. Commun.* 35 (2014) 113–122, <https://doi.org/10.1097/mnm.0000000000000048>.
- [8] A. Sindoni, F. Minutoli, A. Pontoriero, G. Iati, S. Baldari, S. Pergolizzi, Usefulness of four dimensional (4D) PET/CT imaging in the evaluation of thoracic lesions and in radiotherapy planning: Review of the literature, *Lung Cancer* 96 (2016) 78–86, <https://doi.org/10.1016/j.lungcan.2016.03.019>.
- [9] F. Kang, W. Mu, J. Gong, S. Wang, G. Li, G. Li, et al., Integrating manual diagnosis into radiomics for reducing the false positive rate of (18)F-FDG PET/CT diagnosis in patients with suspected lung cancer, *Eur. J. Nucl. Med. Mol. Imaging* 46 (2019) 2770–2779, <https://doi.org/10.1007/s00259-019-04418-0>.
- [10] W. Grootjans, F. Tixier, C.S. van der Vos, D. Vriens, C.C. Le Rest, J. Bussink, et al., The impact of optimal respiratory gating and image noise on evaluation of intratumor heterogeneity on 18F-FDG PET imaging of lung cancer, *J. Nucl. Med.* 57 (2016) 1692–1698, <https://doi.org/10.2967/jnumed.116.173112>.
- [11] T.C. Huang, K.T. Chou, Y.C. Wang, G. Zhang, Motion freeze for respiration motion correction in PET/CT: a preliminary investigation with lung cancer patient data, *Biomed. Res. Int.* 2014 (2014) 167491, <https://doi.org/10.1155/2014/167491>.
- [12] W. Grootjans, L.F. de Geus-Oei, A.P. Meeuwis, C.S. van der Vos, M. Gotthardt, W. J. Oyen, et al., Amplitude-based optimal respiratory gating in positron emission tomography in patients with primary lung cancer, *Eur. Radiol.* 24 (2014) 3242–3250, <https://doi.org/10.1007/s00330-014-3362-z>.
- [13] K. Farid, X. Poullias, M. Alifano, J.F. Regnard, V. Servois, N. Caillaud-Vigneron, et al., Respiratory-gated imaging in metabolic evaluation of small solitary pulmonary nodules: 18F-FDG PET/CT and correlation with histology, *Nucl. Med. Commun.* 36 (2015) 722–727, <https://doi.org/10.1097/mnm.0000000000000311>.
- [14] J.S. Kim, C.R. Park, S.H. Yoon, J.A. Lee, T.Y. Kim, H.J. Yang, Improvement of image quality using amplitude-based respiratory gating in PET-computed tomography scanning, *Nucl. Med. Commun.* 42 (2021) 553–565, <https://doi.org/10.1097/mnm.0000000000001368>.
- [15] S.A. Nehmeh, Y.E. Erdi, C.C. Ling, K. Rosenzweig, Effect of respiratory gating on quantifying PET images of lung cancer, *J. Nucl. Med.* 43 (2002) 876–881.
- [16] L. Guerra, E. De Ponti, F. Elisei, V. Bettinardi, C. Landoni, M. Picchio, et al., Respiratory gated PET/CT in a European multicentre retrospective study: added diagnostic value in detection and characterization of lung lesions, *Eur. J. Nucl.*

- Med. Mol. Imaging 39 (2012) 1381–1390, <https://doi.org/10.1007/s00259-012-2148-2>.
- [17] R. Minamimoto, T. Mitsumoto, Y. Miyata, F. Sunaoka, M. Morooka, M. Okasaki, et al., Evaluation of a new motion correction algorithm in PET/CT: combining the entire acquired PET data to create a single three-dimensional motion-corrected PET/CT image, *Nucl. Med. Commun.* 37 (2016) 162–170, <https://doi.org/10.1097/MNM.0000000000000423>.
- [18] R. Frood, G. McDermott, A. Scarsbrook, Respiratory-gated PET/CT for pulmonary lesion characterisation-promises and problems, *Br. J. Radiol.* 91 (2018) 20170640, <https://doi.org/10.1259/bjr.20170640>.
- [19] W. van Elmp, J. Hamill, J. Jones, D. De Ruysscher, P. Lambin, M. Ollers, Optimal gating compared to 3D and 4D PET reconstruction for characterization of lung tumours, *Eur. J. Nucl. Med. Mol. Imaging* 38 (2011) 843–855, <https://doi.org/10.1007/s00259-010-1716-6>.
- [20] A. Salavati, S. Borofsky, T.K. Boon-Keng, S. Houshmand, B. Khiewvan, B. Saboury, et al., Application of partial volume effect correction and 4D PET in the quantification of FDG avid lung lesions, *Mol. Imag. Biol.* 17 (2015) 140–148, <https://doi.org/10.1007/s11307-014-0776-6>.
- [21] C.S. van der Vos, A.P.W. Meeuwis, W. Grootjans, L.F. Geus-Oei, E.P. Visser, Improving the spatial alignment in PET/CT Using amplitude-based respiration-gated PET and patient-specific breathing-instructed CT, *J. Nucl. Med. Technol.* 47 (2019) 154–159, <https://doi.org/10.2967/jnmt.118.215970>.
- [22] W. Wen, Y. Piao, D. Xu, X. Li, Prognostic value of MTV and TLG of (18)F-FDG PET in patients with stage I and II non-small-cell lung cancer: a meta-analysis, *Contrast Media Mol. Imaging* 2021 (2021) 7528971, <https://doi.org/10.1155/2021/7528971>.
- [23] J.A. Oliver, M. Budzevich, G.G. Zhang, T.J. Dilling, K. Latifi, E.G. Moros, Variability of image features computed from conventional and respiratory-gated PET/CT images of lung cancer, *Transl. Oncol.* 8 (2015) 524–534, <https://doi.org/10.1016/j.tranon.2015.11.013>.
- [24] S. Yip, K. McCall, M. Aristophanous, A.B. Chen, H.J. Aerts, R. Berbeco, Comparison of texture features derived from static and respiratory-gated PET images in non-small cell lung cancer, *PLoS One* 9 (2014) e115510.
- [25] M. Carles, I. Torres-Espallardo, A. Alberich-Bayarri, C. Olivas, P. Bello, U. Nestle, et al., Evaluation of PET texture features with heterogeneous phantoms: complementarity and effect of motion and segmentation method, *Phys. Med. Biol.* 62 (2017) 652–668, <https://doi.org/10.1088/1361-6560/62/2/652>.
- [26] M. Carles, T. Bach, I. Torres-Espallardo, D. Baltas, U. Nestle, L. Marti-Bonmati, Significance of the impact of motion compensation on the variability of PET image features, *Phys. Med. Biol.* 63 (2018) 065013, <https://doi.org/10.1088/1361-6560/aab180>.
- [27] A.M. Garcia Vicente, A.M. Soriano Castrejon, M.P. Talavera Rubio, A.A. Leon Martin, A.M. Palomar Munoz, J.P. Pilkington Woll, et al., (18)F-FDG PET-CT respiratory gating in characterization of pulmonary lesions: approximation towards clinical indications, *Ann. Nucl. Med.* 24 (2010) 207–214, <https://doi.org/10.1007/s12149-010-0345-2>.
- [28] M. Messerli, V. Liberini, H. Grünig, A. Maurer, S. Skawran, N. Lohaus, et al., Clinical evaluation of data-driven respiratory gating for PET/CT in an oncological cohort of 149 patients: impact on image quality and patient management, *Br. J. Radiol.* 94 (2021) 20201350, <https://doi.org/10.1259/bjr.20201350>.
- [29] A.L. Kesner, J.H. Chung, K.E. Lind, J.J. Kwak, D. Lynch, D. Burckhardt, et al., Validation of software gating: a practical technology for respiratory motion correction in PET, *Radiology* 281 (2016) 239–248, <https://doi.org/10.1148/radiol.2016152105>.
- [30] M. Carles, T. Fechter, G. Radicioni, T. Schimek-Jasch, S. Adebahr, C. Zamboglou, et al., FDG-PET radiomics for response monitoring in non-small-cell lung cancer treated with radiation therapy, *Cancers* 13 (2021), <https://doi.org/10.3390/cancers13040814>.